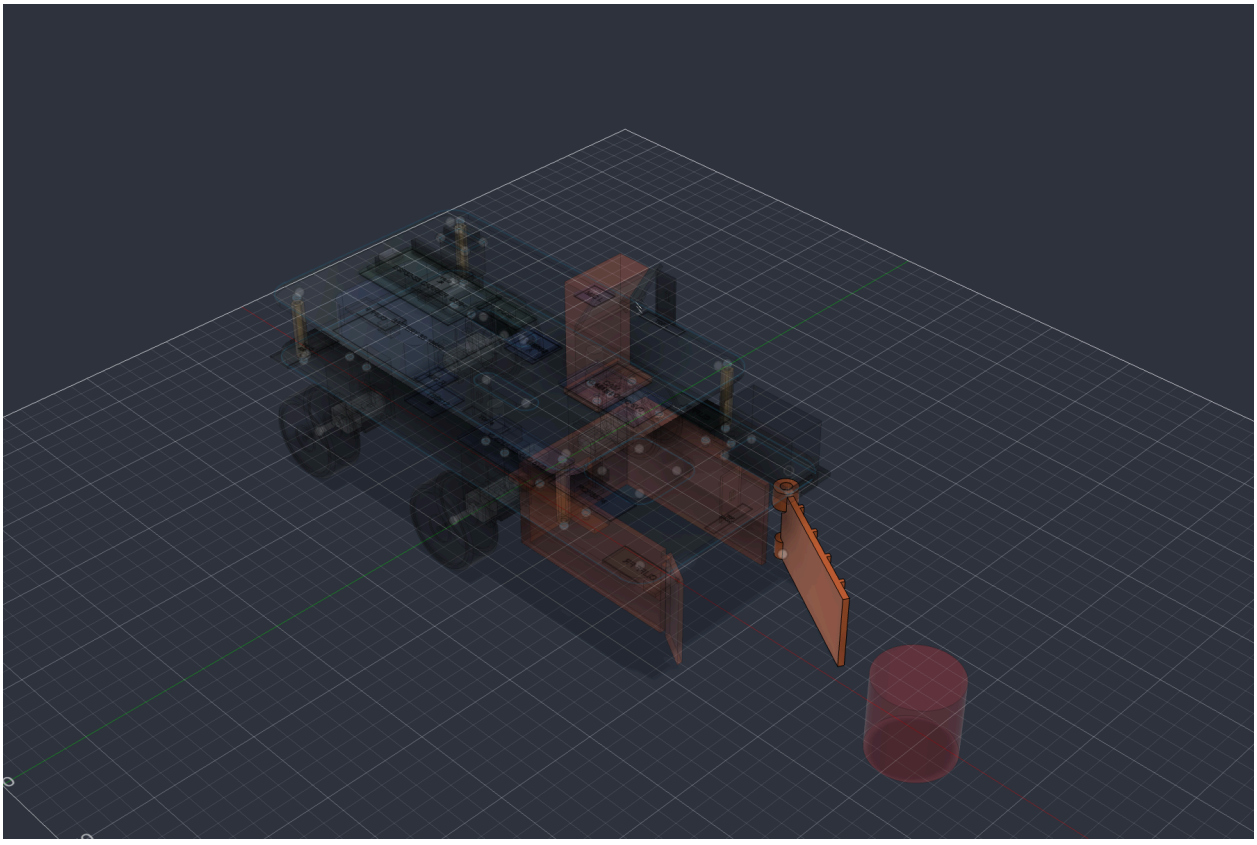
A.E.G.I.S. MK2

Fabrication guide

What to cut, print and prepare, and from which material



Autonomous search-and-rescue rover

TL101 Tinkering Lab, IIT Ropar, AY 2026-27

Item numbers (#) refer to the lines in `to_submit/ItemList_AEGIS_Thursday_1.xlsx`. Build files are in `team_share/Build_AEGIS_Rover/output/` and `team_share/print/`.

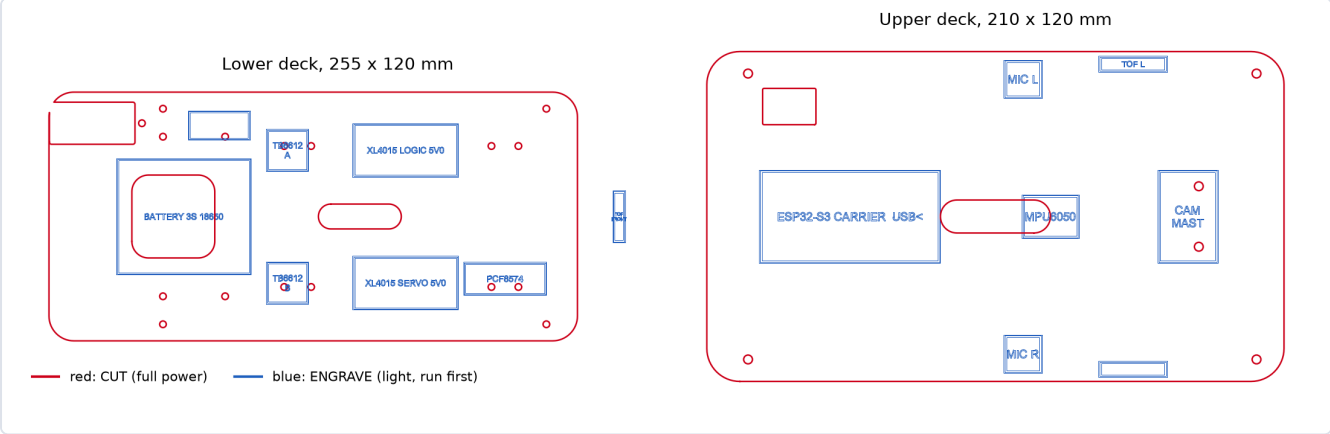
> **Before cutting anything:** rerun both Fusion scripts (`Build_AEGIS_Rover`, `Build_AEGIS_Rover_Detailed`), check the clash list is empty, then regenerate the deck files and print plates: `bash ~/aegis_venv/bin/python team_share/tools/lasercad_dxf.py` `bash ~/aegis_venv/bin/python team_share/tools/make_3mf.py`

1. Summary

What you make	From (item #)	Machine	File
Lower deck, 255 x 120 mm	3 mm clear acrylic sheet (#16)	Laser cutter (LaserCAD)	`lasercad_lower_deck.dxf`
Upper deck, 210 x 120 mm	Same sheet (#16)	Laser cutter (LaserCAD)	`lasercad_upper_deck.dxf`
Cliff-sensor template	Plain paper	Normal printer, 100 % scale	`lasercad_lower_deck_underside.dxf`
Shroud, door, camera mast, 4 motor mounts, 3 ToF brackets	Black PLA+ (#14)	Creality K1 Max	`print/AEGIS_plate1_black_PLA+_rover_parts.3mf`
Victim token, cube decoy, disc decoy	Orange PLA (#39)	Creality K1 Max	`print/AEGIS_plate2_orange_PLA_demo_props.3mf`
ESP32-S3 carrier board, 66 x 34 mm	Prototype PCB 8 x 12 cm (#31)	Hacksaw or score-and-snap, file the edges	Size from this guide
Fuse board, 30 x 14 mm	Offcut of the same PCB (#31)	Same	Size from this guide
Header pins	Male and female strips (#32, #33)	Side cutters	Counts in section 4
Wires	Silicone wire (#24–#30)	Wire strippers	Gauges in `AEGIS_Wiring_Guide.pdf`

Nothing else is cut or printed: wheels, motors, spacers, screws and every electronic board are bought ready-made.

2. Acrylic decks (laser cutter, LaserCAD)

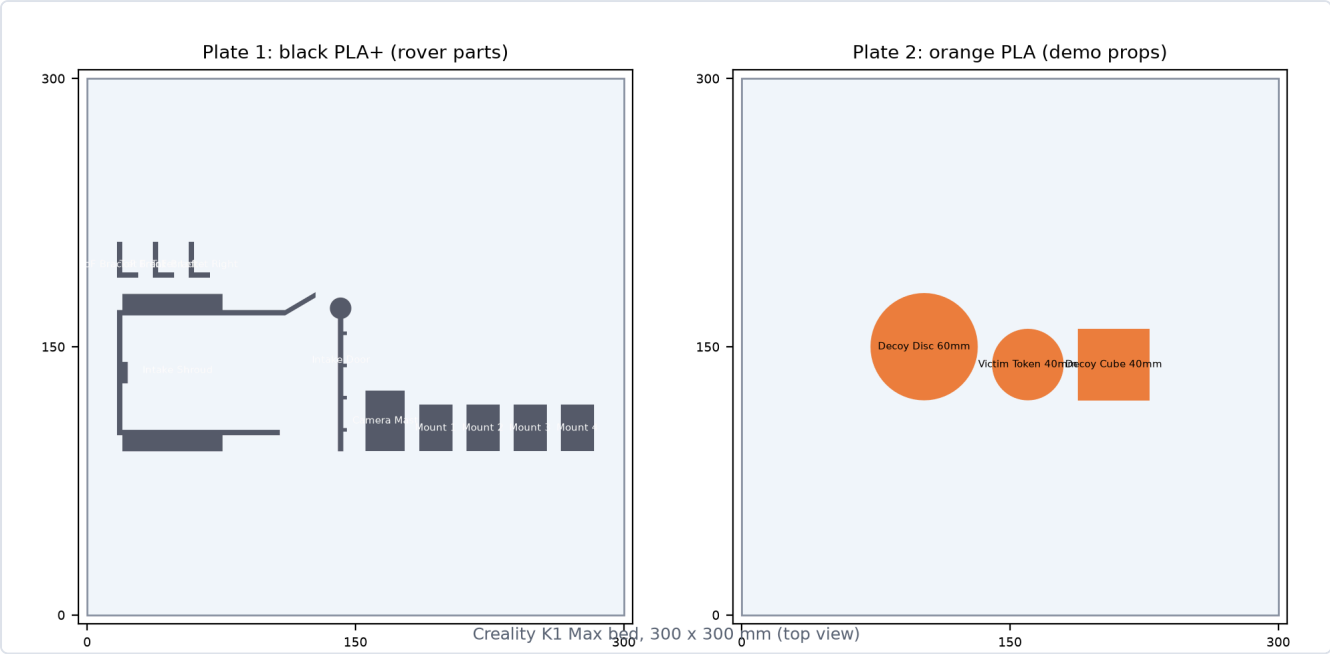


The two decks as LaserCAD will show them (lasercad_*.dxf).

Material: one NEILL 12 x 12 inch (about 305 x 305 mm) clear acrylic sheet, 3 mm (#16). Both decks fit on it.

1. Peel the protective film only after cutting (it stops burn marks), unless the lab says otherwise.
2. In LaserCAD, import `lasercad_lower_deck.dxf` and `lasercad_upper_deck.dxf`. **Choose millimetres** when asked, and check the sizes: 255 x 120 mm and 210 x 120 mm.
3. Place them side by side along the 305 mm edge, about 30 mm apart (some labels reach a few mm past the deck outline). They need roughly 305 x 255 mm.
4. Layers: - **Blue (ENGRAVE):** placement marks and labels. Low-power scan or light engrave. Run first. - **Red (CUT):** outline, holes, slots. Full cut for 3 mm acrylic. Run second. Ask the lab staff for their machine's power and speed for 3 mm acrylic; do a 20 mm test square on a corner of the sheet first.
5. Check after cutting: M3 screws (3.2 mm holes) pass freely; the servo pocket fits the MG946R body; the power-switch cutout (19.4 x 13.2 mm) fits the switch. Measure the switch before cutting if it has arrived.
6. Do **not** laser the underside file. Print `lasercad_lower_deck_underside.dxf` on paper at 100 % (check a dimension with a ruler) and use it as a template for the five cliff sensors under the lower deck.

3. 3D-printed parts (Creality K1 Max, Creality Print)



Print plates as laid out in the two 3MF files.

Plate	Filament	Parts
1	Black PLA+ (#14)	Intake shroud, intake door, camera mast, 4 motor mounts, 3 ToF L-brackets
2	Orange PLA (#39)	Victim token (40 x 40 mm), cube decoy (40 mm), disc decoy (60 x 12 mm)

Settings: K1 Max profile, textured PEI plate, 0.2 mm layers, 3 walls, 25 % gyroid infill; 40 % for the motor mounts, camera mast and brackets. Brim on the shroud. No supports needed: every part is already oriented flat.

Order: 1. **Fit test first:** print one motor mount and the door alone. Check the N20 motor slides into the mount (0.2 mm clearance per side; sand lightly if tight) and the servo spline presses into the door socket (warm the spline; if it is loose, glue the servo horn instead). 2. Print the rest of plate 1. 3. Print plate 2 any time. Paint or tape a white band around the middle of the token.

Filament use: at most about 160 g black and 180 g orange (solid-part estimate; less with infill), from 1 kg spools.

4. Electronics prep (bench, soldering iron)

Prototype PCB (#31), 8 x 12 cm: - Carrier board: 66 x 34 mm. Holds two 22-pin female sockets for the ESP32-S3 (rows 22.86 mm apart, 2.54 mm pitch), the two 2.2 kΩ I2C pull-ups and the 100 kΩ / 20 kΩ battery divider. - Fuse board: 30 x 14 mm. Holds the BLX-A fuse holder (#22). - Score both sides along a hole row with a knife and snap, or use a hacksaw; file the edges smooth.

Female header strips (#32), 4 x 1x20: each ESP32-S3 row needs 22 sockets. Use one full strip plus 2 sockets cut from a second strip, per row. Cutting a female strip sacrifices the socket you cut through, so cut on the 3rd socket of the spare strip.

Male header strips (#33), 3 x 1x40: break off by hand:

Board	Pins
VL53L0X x 3	6 each = 18
MPU-6050	8
INMP441 x 2	6 each = 12
TB6612FNG x 2	16 each = 32
Total	70 (two strips; the third is spare)

Wire (#24–#30): cut to length as you route, following `AEGIS_Wiring_Guide.pdf`:

Wire	Use	You have
18 AWG red / black	Battery, fuse, switch, buck and driver inputs	1 m each
22 AWG 3-core	5 V rails, servo, motor leads	2 m
26 AWG red / black / white / green	3.3 V, grounds, signals	3 m each

Slide heat-shrink (#35) on before soldering every power joint.

Foam tape (#38): two layers on the camera pad of the mast; one layer under each sensor board, the ToF brackets, the mic boards and the PCF8574.

Cable ties (#36): one per motor through the channel in its mount; the rest for the harness and to strap the battery to the lower deck.

5. Order of work

1. Fusion rerun, then regenerate the DXF and 3MF files (top of this guide).
2. Fit-test prints (motor mount, door), then plate 1 and plate 2.
3. Laser-cut the decks.
4. Cut and solder the carrier board and fuse board; solder headers on the modules.
5. Assemble and wire with `AEGIS_Wiring_Guide.pdf`. Set both XL4015 outputs to 5.0 V with a multimeter **before** connecting any board.